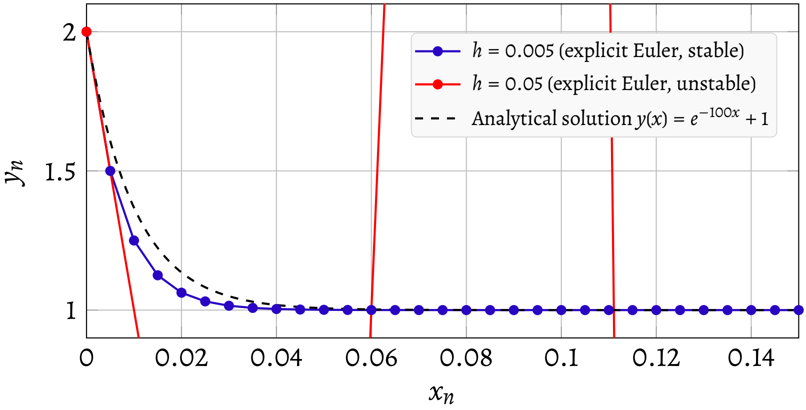

Figure 19 – Numerical stiffness in the equation $y' = -100y + 100$, $y(0) = 2$. For a small step ($h = 0.005$), the explicit Euler method follows the stable analytical solution and converges rapidly to $y = 1$. For a larger step ($h = 0.05$), the same method becomes numerically unstable and the solution blows up, despite the true dynamics being perfectly stable.